

Case Study: Solving a Real-World Problem Using Creative Thinking Techniques

1. Introduction

Real-world problems often require structured thinking and creativity to develop effective solutions. This case study focuses on identifying a real-world problem, generating multiple solutions using creative thinking techniques, and evaluating those solutions using established decision-making frameworks.

For this study, the selected problem is **traffic congestion in urban areas during peak hours**.

2. Problem Identification

Problem Statement

Traffic congestion in urban cities causes delays, increased fuel consumption, pollution, and reduced productivity.

Key Causes

- Increasing number of private vehicles
- Limited road infrastructure
- Poor public transport planning
- Inefficient traffic signal systems

Impact

- Economic losses due to delays
- Environmental pollution

- Stress and reduced quality of life

3. Creative Thinking Techniques for Solution Generation

To generate possible solutions, two creative thinking techniques were used:

1. Brainstorming

Brainstorming allows multiple ideas to be generated without immediate criticism.

Ideas generated:

- Expand public transportation systems
- Promote carpooling
- Introduce smart traffic signal systems
- Implement congestion pricing
- Encourage remote work options

2. SCAMPER Technique

SCAMPER stands for **Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse**.

Using SCAMPER produced additional ideas:

- Substitute fuel vehicles with electric buses
- Combine metro and bus ticketing systems
- Modify traffic signals using AI
- Reverse traffic lanes during peak hours

4. Proposed Solutions

From the ideas generated, three major solutions were selected:

1. **Smart AI-based traffic signal systems**
2. **Carpooling and ride-sharing incentives**
3. **Improved public transportation infrastructure**

5. Evaluation Framework

To evaluate the solutions, a **Decision Matrix Framework** was used with the following criteria:

- Cost
- Implementation Time
- Effectiveness
- Sustainability
- Public Acceptance

Solution	Cost	Implementation Time	Effectiveness	Sustainability	Public Acceptance
AI Traffic Signals	Medium	Medium	High	High	High
Carpool Incentives	Low	Low	Medium	Medium	Medium
Public Transport Expansion	High	High	Very High	Very High	High

6. Analysis of Solutions

AI Traffic Signal Systems

- Adjust signal timing dynamically
- Reduce waiting time at intersections
- Moderate cost and high effectiveness

Carpooling Incentives

- Encourages shared rides
- Low cost but depends on user participation

Public Transportation Expansion

- Most effective long-term solution
- High cost and long implementation period

7. Recommended Solution

Based on the evaluation, **AI-based traffic signal systems** are the most practical short-term solution because they:

- Reduce congestion quickly
- Require moderate investment
- Can be implemented using existing infrastructure

However, **public transportation expansion** remains the best **long-term solution**.

8. Conclusion

This case study demonstrated the process of solving a real-world problem using structured problem-solving techniques. By applying brainstorming and SCAMPER methods, multiple solutions were generated and then evaluated using a decision matrix. The analysis identified AI-based traffic signals as the most feasible immediate solution, while long-term improvements require investment in public transportation infrastructure.